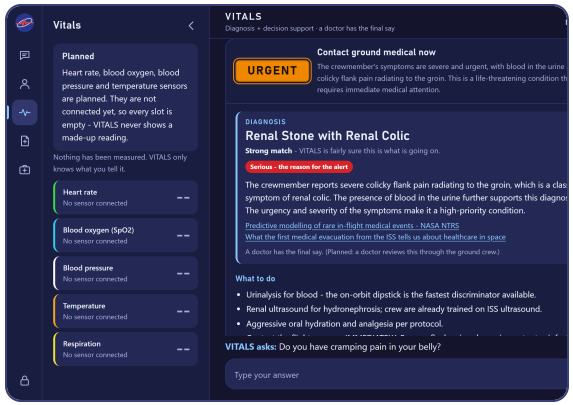
BUILT TODAY · PLANNED NEXT

One screen working. The rest is the plan.

- **WORKS** Astronaut screen: PIN login, emergency access, chat, answer card, corrections, profile
- **WORKS** Reasoning over 35 cited spaceflight conditions with a safety backstop
- **WORKS** Every answer saved as a case, every step logged
- **PLANNED** Ground crew screen — reviews every case, decides what a doctor sees
- **PLANNED** Doctor screen — agrees, disagrees, or gives their own read
- **PLANNED** Sensors: heart rate, blood oxygen, blood pressure, temperature
- **PLANNED** Jetson server + three Raspberry Pi screens · voice input



Planned parts say so in the app. Sensor slots stay empty — never a made-up reading.

TEAM TETHER

Who we are

SCHOOL	[School name] · No. [school number]
CITY	[City], Texas
TEAM	TETHER — Joshua Collado (project lead, code) · Joaquin (medical research, code) · Cruz (interface, user testing)
TEACHER	Ms. Sylvia Odoms
MENTOR	Michael Hayes, Voyager Technologies
NASA PROJECT	Medical Diagnostic System with Machine Learning & Artificial Intelligence (From a non-Medical Person's Perspective)
OUR PROJECT	VITALS

Website QR
python
pdr/website/make_qr.py

WEBSITE

Specs QR
same, with

#specs

TECHNICAL SPECS

VITALS is a student prototype for diagnosis and decision support — a doctor has the final say. Not a medical device. NASA HUNCH 2026–27 · Software & Technology Hardware Engineering plus Robotics.

VITALS

When the doctor is 20 minutes away, the crew still needs an answer now.

TEAM TETHER

NASA HUNCH 2026–27

An onboard medical assistant for long-duration Moon and Mars missions

THE PROBLEM

No doctor on board. Help is up to 40 minutes away.

On Mars, a message to the flight surgeon takes 4 to 40 minutes each way, and NASA plans for blackouts of up to two weeks. Most crews will have no physician — at best a crewmember with paramedic training. When someone gets sick or hurt, they have to answer three questions fast:

1. **What could this be?**
2. **How serious is it** — do we call the ground now?
3. **What do we do** in the meantime?

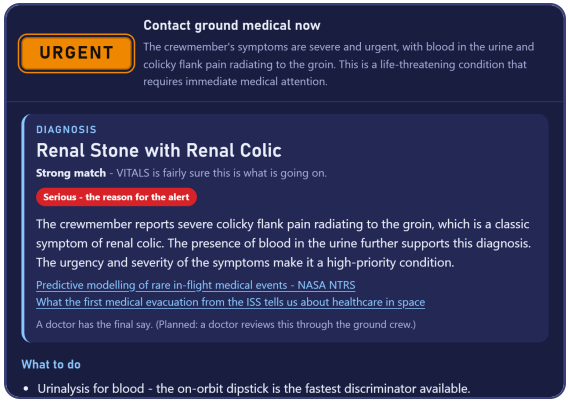
> **VITALS** is an onboard medical assistant a non-medical crewmember can talk to in plain words. It runs with no internet, it knows spaceflight medicine, it puts urgency first — and it shows the source behind every answer.

Earth symptom checkers assume Earth. In space, a stuffy nose on day 2 is fluid shift, not a cold — and joint pain after a spacewalk is decompression sickness until proven otherwise.

HOW IT WORKS

Grounded AI, with a safety net

ASK	Crewmember types how they feel
▼	
SCOPE	Not medical? A plain reply — the AI never runs
▼	
READ	Pull the symptoms out of the words; crew can fix mistakes
▼	
RETRIEVE	Pick the cited research pages that fit
▼	
REASON	Local AI reads *only* those pages and answers
▼	
CHECK	Drop invented conditions · sources from our knowledge base · rules engine can always escalate
▼	
ANSWER	Urgency first, then what it could be, why, and what to do



A real answer from the prototype: side pain in waves with blood in the urine → URGENT, kidney stone, with sources.

WHAT MAKES IT DIFFERENT

Top features

- **OFFLINE** — Model built into the system — no internet, no cloud, no data leaves the habitat
- **SOURCED** — 35 spaceflight conditions from NASA and peer-reviewed research; the AI can't invent a citation
- **SAFE** — A rules engine can raise the alarm even when the AI doesn't
- **HONEST** — No made-up vital signs, no fake percentages
- **FIXABLE** — A *Wrong?* button on everything it understood

35
cited conditions

262
automated tests

18/19
urgent complaints escalated (Oct 17) — the miss is our first fix

6/6
basic missions, every step logged

Technical specs

AI	Local language model (llama3.2, 3B) via Ollama, temperature 0
SOFTWARE	Python · stdlib web server · Docker
HARDWARE	NVIDIA Jetson Orin Nano 8 GB · Raspberry Pi screens **PLANNED**